

Motherboard

Motherboard

- Motherboard is the PC's center of activity.
- All devices in a computer are in some way or another are connected to motherboard.
- Motherboard hosts the largest single collection of chips of any pc component. It has a grid of wires that link all components making it possible to communicate with each other.
- Open circuit paths called traces that resemble complicated road map carry signals (data) and voltage across the motherboard.
- Therefore, it is not surprising if motherboard is data and power infrastructure of the entire computer. It defines the computer's limit of speed, memory, and expandability.

Chipsets

- A motherboard comes with a variety of support chips soldered in place. These chipsets are designed to work with CPU.
- These chips are highly complex and coordinated ICs that help CPU manage and control the computer system.
- When replacing CPU, you must make sure the CPU works (compatible) with the chipset supported by the motherboard. Otherwise the computer will not work.

Motherboard

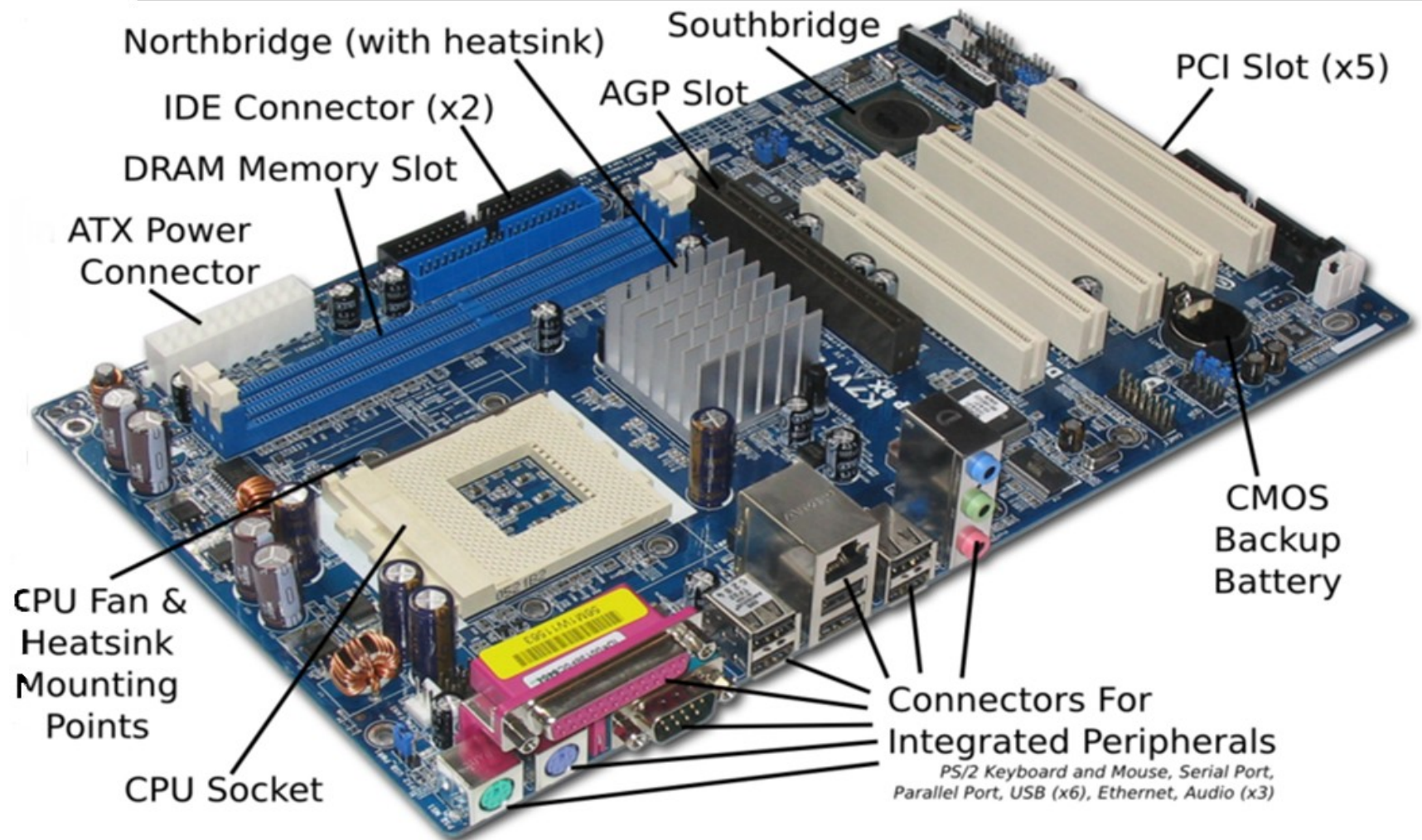


Fig 2.4 typical components on the motherboard

- A basic chipset consists of:
 - 1) Bus controller
 - 2) Memory controller
 - 3) Data and address buffer
 - 4) Peripheral controller

- The number of chips a motherboard differs from one type of motherboard to another.

Types of motherboard

- Motherboards come in several standardized configurations called form-factors.
- Form-factors define the size, location of expansion slot, what sorts of power supply it uses, its physical organization, etc of the motherboard.
- The following are some of the common form-factors:
 - 1) Obsolete form factors:
 - Baby AT
 - Full-size AT
 - LPX
 - 2) Modern form factors:
 - ATX
 - Micro-ATX
 - Flex ATX
 - NLX
 - WTX
 - BTX
 - 3) Proprietary designs.
E.g. Compaq

Baby AT

- The first popular pc motherboard was the original IBM PC released in 1981.

- IBM released a motherboard called XT in 1983. This motherboard became very popular. Based on this motherboard, IBM released AT motherboard in 1984.

- Baby AT motherboards are 8.5x13 inches. They are distinguished by their shape and usually by the presence of single full sized keyboard connector soldered onto the motherboard.

- All other devices are connected to the motherboard via cards and expansion slots or cables. The serial and parallel ports are attached using cables that go between the physical connectors mounted on the case and pin headers located on the motherboard.

Full Sized AT

- Full Size AT is larger than Baby AT. It is 12 inches wide by 13.8 inches long.
- These motherboards fit into towers only. They do not fit into Baby AT or mini tower cases. This kind of motherboard is common in older machines, 386 class or earlier.
- Both Full Size AT and Baby AT form factors put the processor sockets (slots) and memory sockets at the front side of the motherboard.

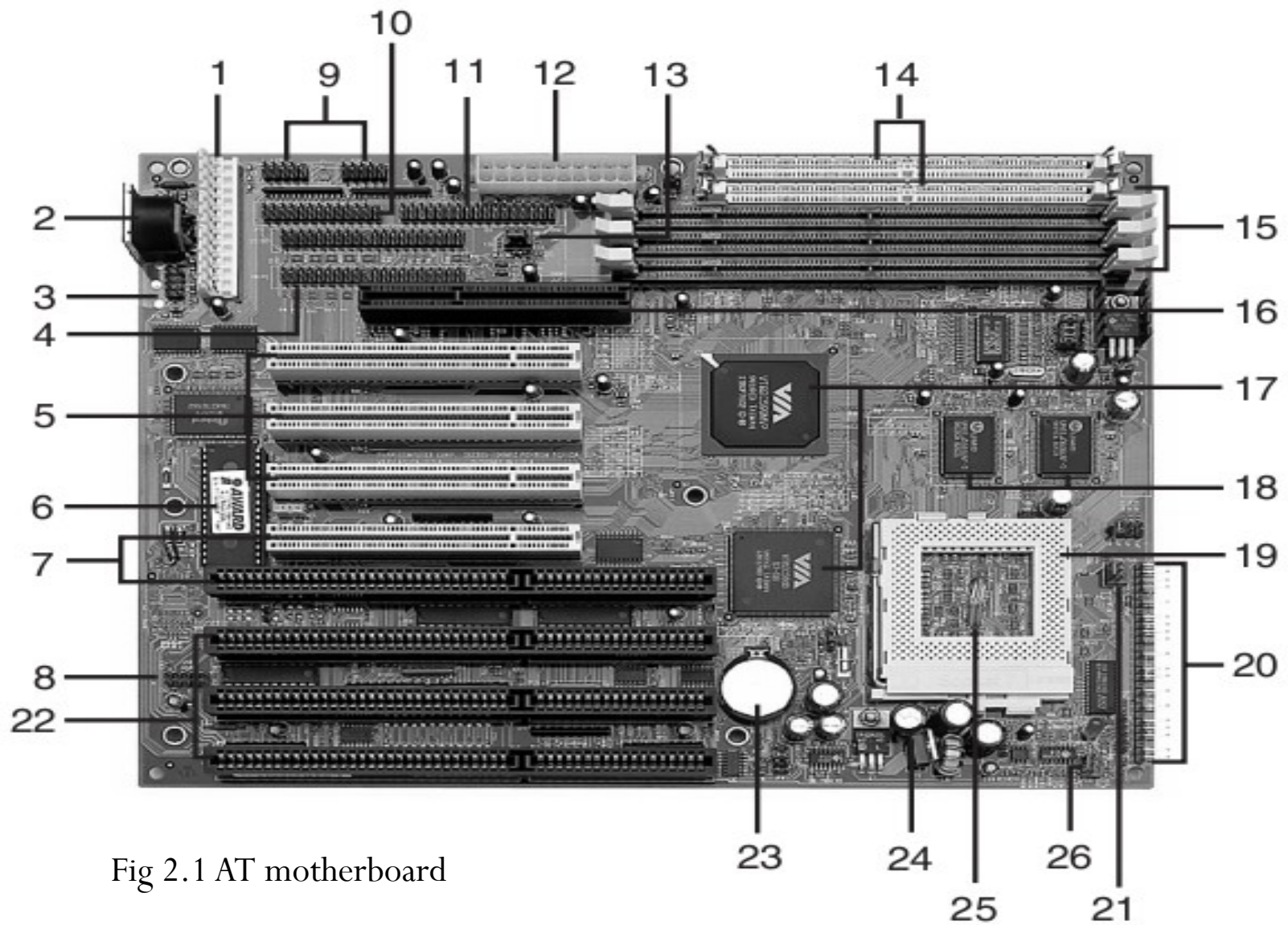


Fig 2.1 AT motherboard

Motherboard ...

- AT motherboards get power through P8 and P9 connectors.
- Each of them has two black wires for grounding and these powers must be kept together when the connectors are plugged in i.e. all the four black wires sit in the center when plugged unless stated on motherboard manual.

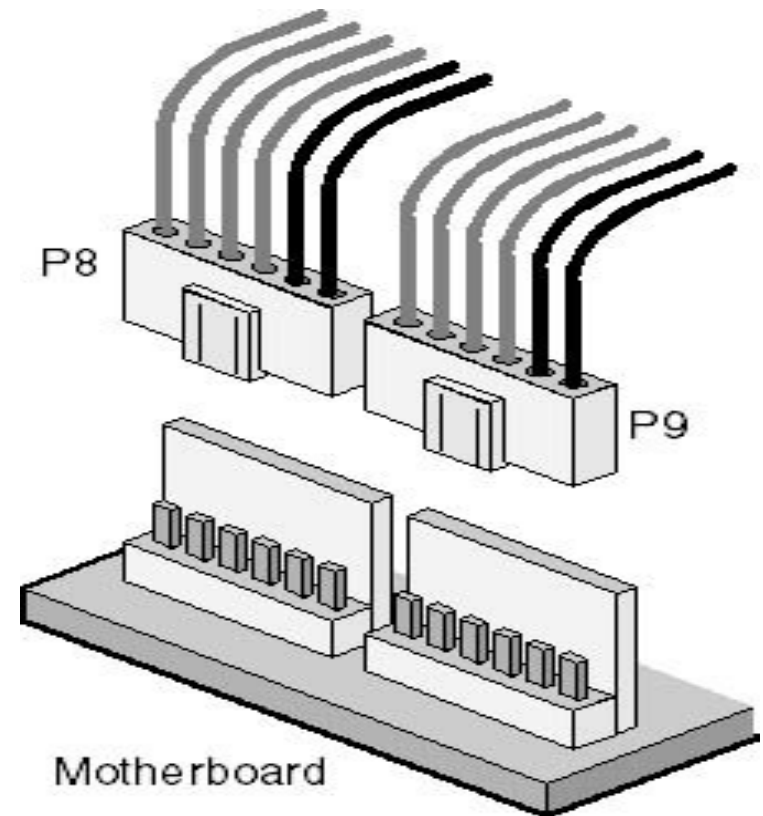


Fig 2.2 P8 and P9 connectors

Motherboard ...

ATX Motherboard (AT Extended)

- ATX form factor was introduced in 1996 by Intel. It measures 12 by 9.6 inches. Although it is approximately the same size as Baby AT, the ATX motherboard is rotated 90 degrees; the processor sits near the back of the board.
- In ATX motherboards, the connectors COM1, COM2, LPT, keyboard, mouse, and USB are directly mounted on the motherboard. This motherboard requires a specifically designed chassis with an I/O access opening. In addition, the memory and expansion slots are located at 90 degrees to each other.
- ATX motherboard is more intelligent than ordinary type. It has advanced control facilities where BIOS program checks the CPU temperature and voltage, and the cooling fan RPM(rotation per minute), etc. If overheating occurs, the PC will shutdown automatically.

Motherboard ...

Fig ATX motherboard



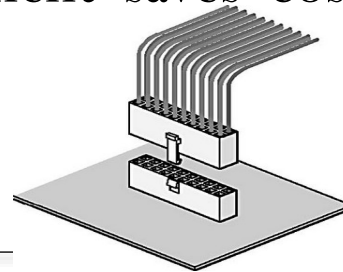
Motherboard ...

- Some of the advantages of ATX over AT are:
 - 1) ***Soft power support:*** the ATX power supply is turned on and off using signaling from the motherboard, not a physical toggle switch. This allows the PC to be turned on or off under software control, allowing improved power management.
 - 2) ***Reduced expansion card interference:*** the CPU and memory modules are relocated so they can't interfere with any bus expansion cards. This makes upgrade easy because removing any installed cards is not necessary.
 - 3) ***Reduced drive bay interference:*** since the board is essentially 90 degrees from AT style, there is much less overlap between the board and drive bays. This means easier to access the motherboard and fewer cooling problem.

Motherboard ...

- 4) ***Better Power Connector:*** ATX uses a single 20-pin connector instead of two confusing connectors. You don't have the risk of blowing up your pc motherboard by connecting the power cable backwards.
- 5) ***Better air flow(cooling effect):*** The ATX power is intended to blow air into the case rather than out of it. This cuts down dust. Furthermore, processor sockets and RAM are located next to power supply. Because of this the power supply fan can cool CPU heat sink. Current power supply fans blow air into or out of the case.
- 6) ***Integrated I/O connectors:*** AT motherboard uses pins and cables. In ATX, the port is directly mounted on the motherboard. This improvement saves cost, installation time, and improved reliability.

Fig 4 ATX power supply



LPX

- LPX was developed by Western Digital in 1987. LP stands for Low Profile. It is called so because it allows small cases and overall smaller system over AT.
- The primary goal of LPX is to reduce space usage and cost.
- It the most noticeable is that expansion slots are mounted on a bus raiser card that plugs into the motherboard.
- Expansion cards must plug sideways into raiser card. This allows for low profile case design.
- The cards will be placed parallel to motherboard, not perpendicular to motherboard as in AT or ATX systems.

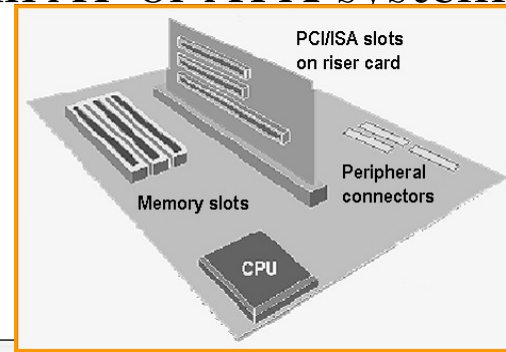


Fig 5 LPX motherboard

LPX ...

- LPX motherboards often come with video display adapter cards built into the motherboard. They also have serial, parallel, and mouse connectors attached to them like ATX.
- Disadvantage:
 - Lack of standardization
 - Poor expandability (limited to two or three expansion slots)
 - Poor upgradeability
 - Poor cooling
- Identification: check if expansion slot is parallel to motherboard. If so, it is either LPX or NLX.

NLX Motherboard

- NLX stands for New Low Profile form factor. It was introduced by Intel in 1996. It is designed to replace the non-standard LPX design used in the previous Low Profile systems.
- NLX still uses the same general design as LPX:
- Smaller motherboard
- Raiser card for expansion cards

NLX ...

- Characteristics of NLX board:
- The motherboard plugs into the riser, unlike LPX where the card plugs into the motherboard. This means motherboard can be removed from case without affecting the raiser card.
- The motherboard has literally no internal cables or connectors attached to it.
- All devices that plug into motherboard such as drive cables, power supply, front panel light, switch connectors, etc. plug into the raiser instead.
- No unplugging is needed to remove the motherboard except unplugging the motherboard from the raiser card.

The NLX Board

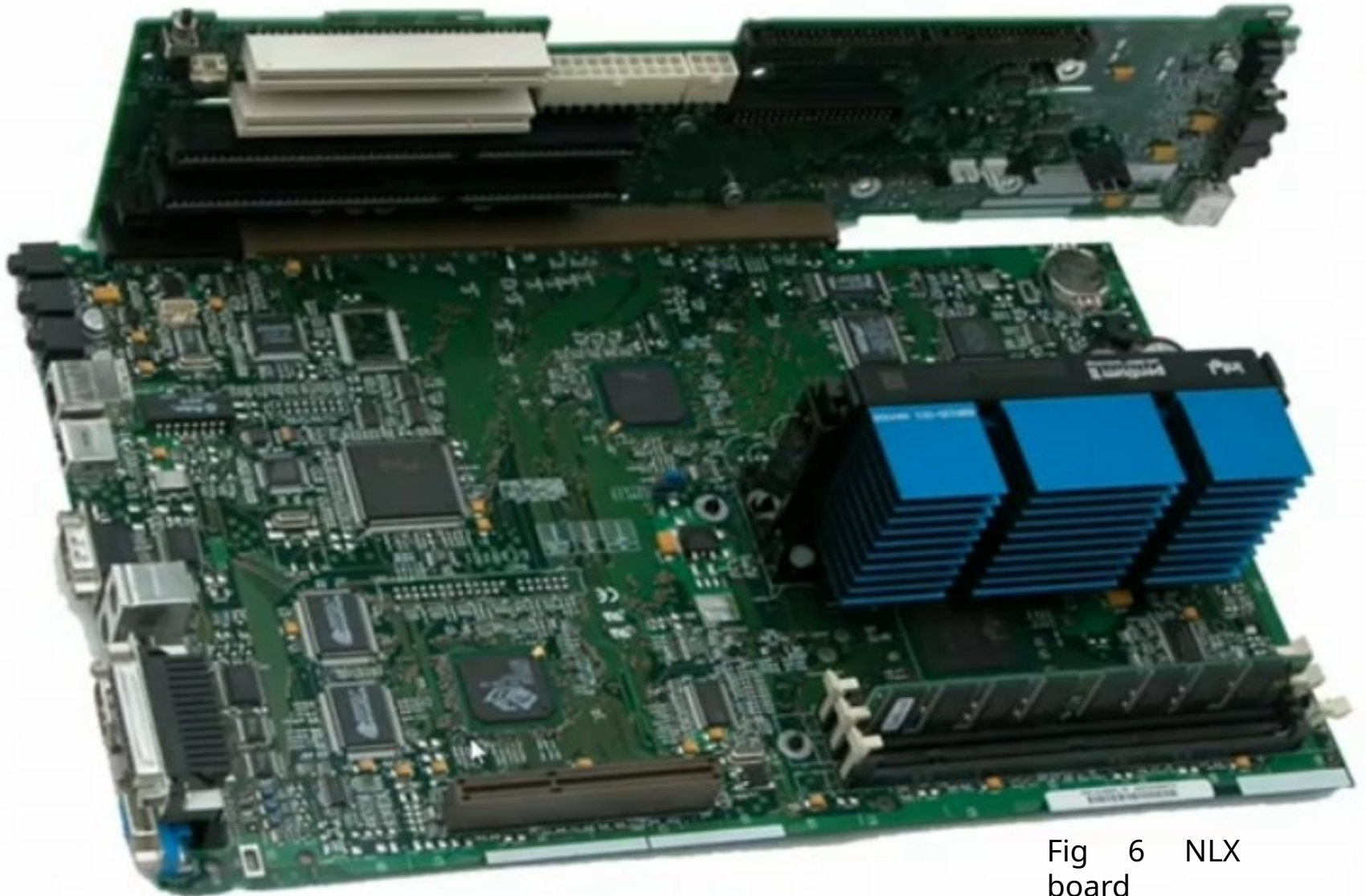


Fig 6 NLX
board

- Advantages over NLX:
 - Revised design to support larger memory modules and modern DIMM memory packaging
 - Support for newest processor technologies including the new Pentium II using SEC packaging
 - Support for AGP cards, USB, etc-newer technologies
 - More optimal location of CPU on the motherboard. This allows easier access and better cooling.
 - Enhanced design such as motherboard can slide in and out of the system case easily

BTX

- BTX stands for Balanced Technology Extended.
- BTX has smaller profile i.e. lower but wider.
- It is created to dissipate heat in a better way than ATX.
- It uses straight lines to allow better air flow to achieve better cooling. It is no longer in production.
- However, many manufacturers use the benefits of BTX design.

BTX ...

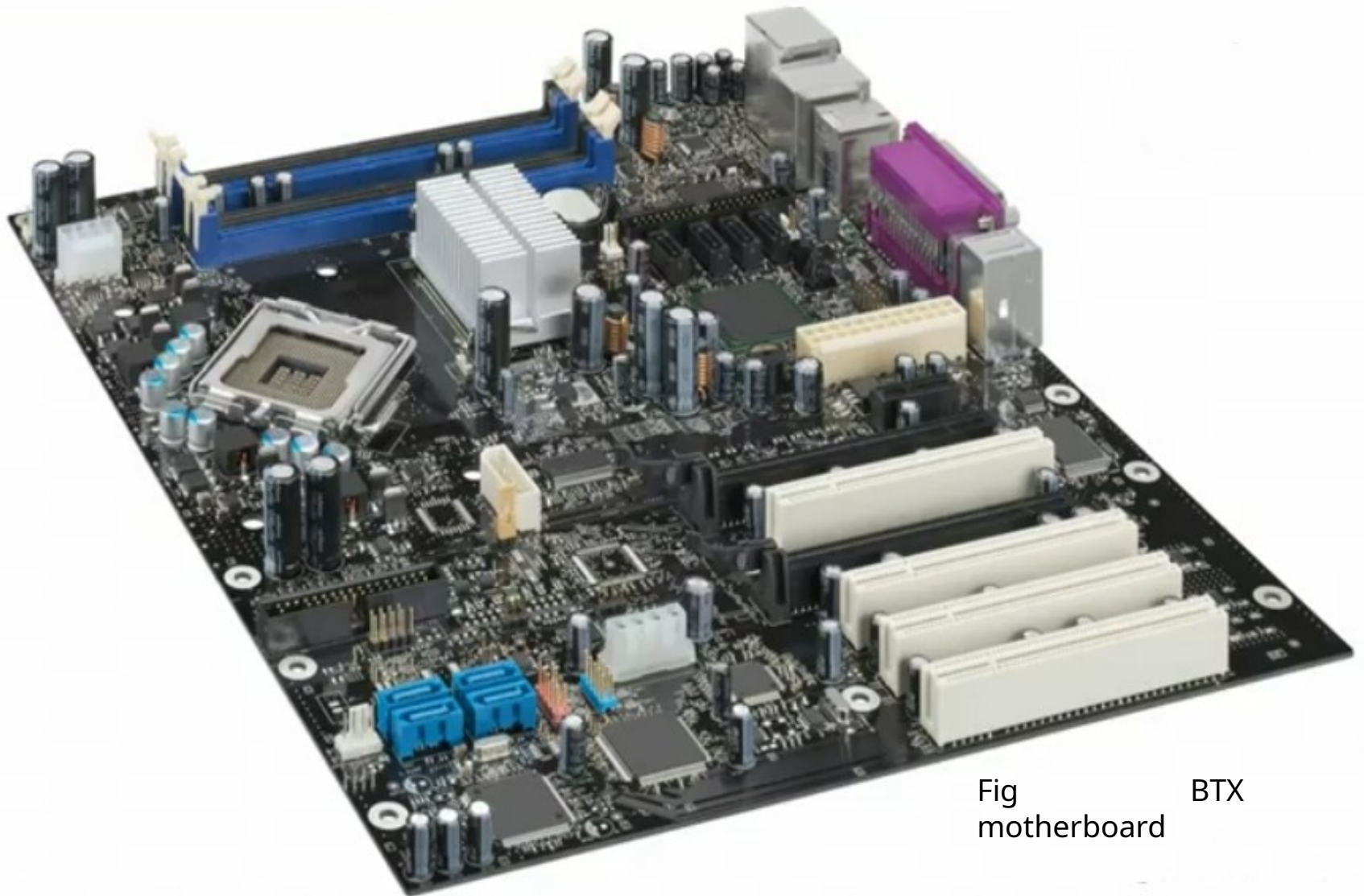


Fig BTX
motherboard

Chipsets

- Chipsets are a bunch of intelligent controller chips, which are found on any system board.
- They are closely tied to the CPU, and they help the CPU control the system.
- Without chipset, neither RAM nor I/O buses like AGP, PCI, ISA could function together with CPU.
- Chipset represents the connection between the CPU and everything else.
- The processor can't talk to the memory, adapter boards, devices, and so on without going through chipset.
- Since chipset controls the interface or connection between the processor and everything else, they end up dictating what type of processor you have, how fast it will run, how fast buses run, the speed, type and amount of memory you can use, etc.

Chipsets ...

- New chips often enable:
 - Higher speed on one or more buses
 - Utilization of new facilities like new RAM types, new buses, improved EIDE, etc.
- There are several suppliers of Pentium chipsets:
 - Intel
 - SIS(Silicon Integrated System)
 - Opti
 - Via
 - Ali (Acer Laboratories, inc)

Chipsets ...

- **Chipset Evolution**
- When the first motherboard was introduced by IBM, they used several discrete chips:
 - Math coprocessor
 - Clock generator
 - Bus controller
 - System timer
 - Interrupt and DMA controllers
 - CMOS RAM and clock
 - Keyboard Controller

Chipsets ...

- In 1986, a company called Chips and Technologies introduced revolutionary component called the 82C206.
- This was a single chip that integrated the functions of Clock Generator, Bus Controller, System Timer, Dual Interrupt Controllers, dual DMA Controllers, and the CMOS/Clock Chip.
- Virtually all the major chips on the motherboard could now be replaced by a single chip. This idea was copied by other manufacturers.

- **Intel Chipsets**

- Intel joined the chipset market in 1989 by introducing the 420 series chipset along with 486 processor. Intel produced CPU before 1989.
- For new CPUs produced by Intel to be on market, compatible chips are necessary. Before chipsets are produced for new Intel CPU it takes as long as a year.
- This means Intel could not sell its CPU as needed. Because of this, Intel started to produce chipset for its CPU.

- Intel started a pattern of number its chipsets:

Chipset number	Processor family
420XX	P4(486)
430XX	P5(Pentium)
440XX	P6 (Pentium pro/II/III
450XX	P6 server(Pentium pro/PII/PIII Xeon)
8XX	P6(PII/III/IV) hub architecture
9XX	P7(Pentium IV) hub architecture

- Intel used two distinct chipset architectures:
 - North/south bridge architecture
 - Hub architecture

- **1) North-South Bridge Architecture**
- Most of earlier Intel chipsets and all non-Intel chipsets are broken down into a multi-tiered architecture. This includes what is referred to as north bridge, south bridge, and super I/O chip.
- **North Bridge:**
- The North Bridge handles communication between the CPU, RAM, AGP on PCI Express and South Bridge. It handles communication with faster devices. Because different processors and RAM requires different signaling, North Bridge will typically work with one or two classes of CPU and generally one type of RAM. There are a few chipsets that support two types of RAM. This occurs when there is a shift to new standard.
- **South Bridge:**
- Because the south bridge is further removed from CPU, it is given the responsibility for slower devices on a typical micro computer. A particular South Bridge will usually work with different North Bridges but these two chips must be designed to work together.

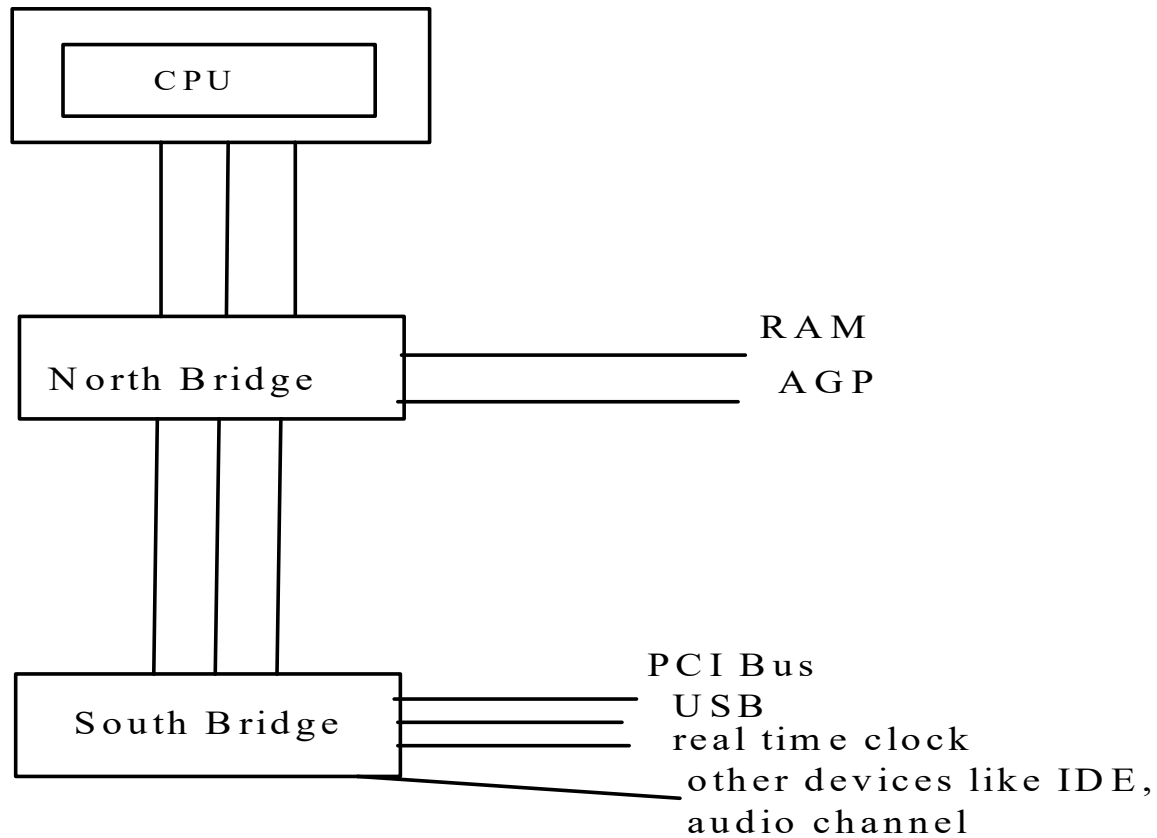


Fig 7 north and south
bridge

- Traditionally the interface between North Bridge and South Bridge was the PCI Bus. However, this created bottleneck in performance. Most current chipsets use a different interface with higher performance.
- The functionality found on contemporary South Bridge includes:
 - PCI bus
 - ISA bus
 - Interrupt controller
 - DMA controller
 - IDE controller(SATA or PATA)
 - Real time clock

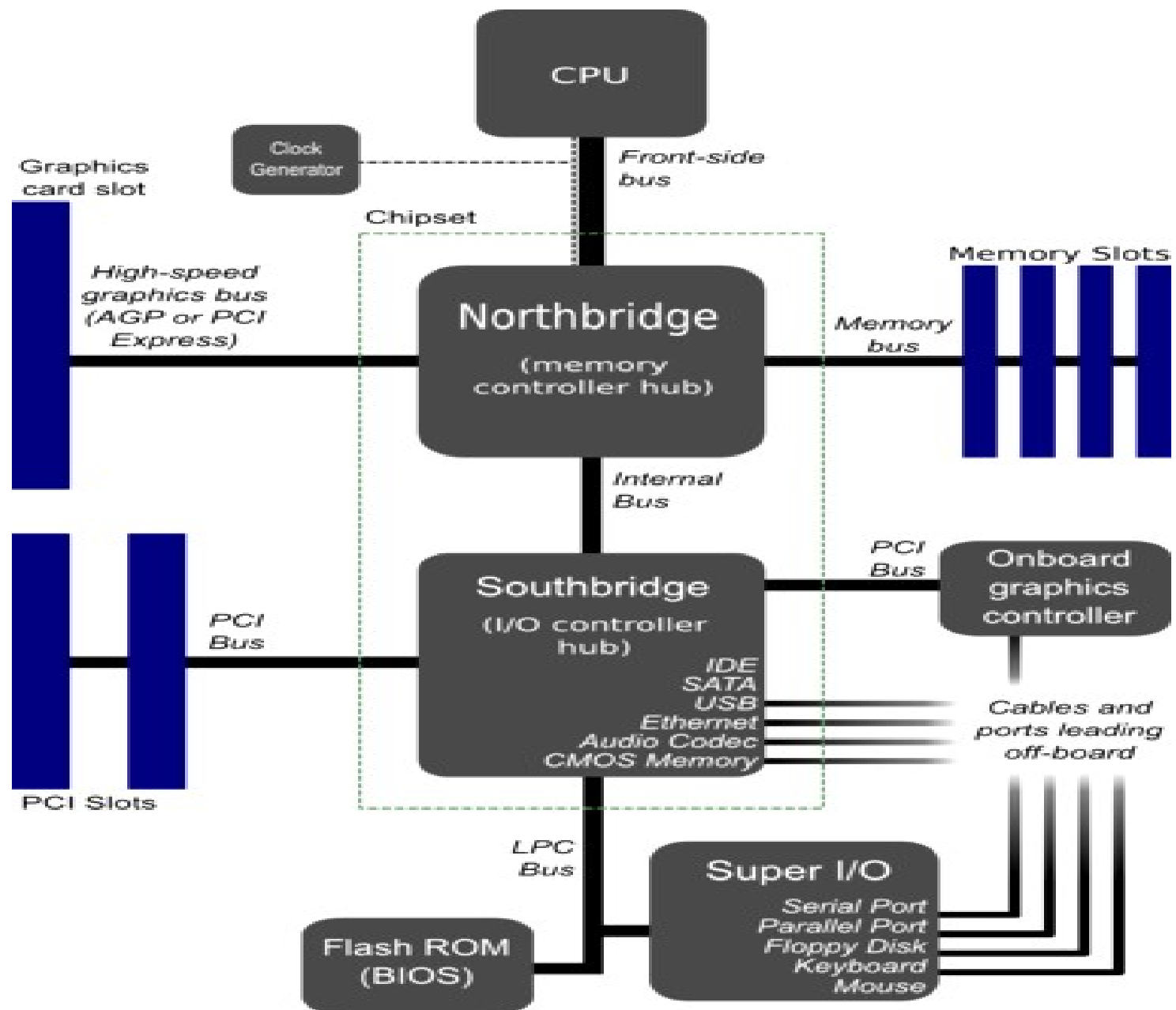


Fig motherboard chipsets and their interconnection

Super I/O Chipset:

- The major function of Super I/O chips are:
 - a) Serial Port controller: super I/O chip controls the serial ports
 - b) Parallel port control: it provides the circuitry to drive the parallel port. This includes support for newer parallel port types.
 - c) Floppy disk control: support for floppy disk drives is provided by the super I/O chip
 - d) Keyboard and mouse controller

2) Hub Architecture

- The newer 800 and 900 series chip uses a hub architecture where the former north bridge is now called Memory Controller Hub and the former South Bridge is called I/O Controller Hub(ICH).
- They are connected via 4X66MHZ hub interface which is twice as fast as PCI and does not share or steal PCI bus bandwidth. This improves performance of all PCI connected devices because the PCI is not involved in these transactions.
- Hub interface:
 - Allows for much greater throughput for PCI devices
 - Allows greater throughput for devices directly connected to the I/O controller hub.
 - Economical

Chipset Generation

Fifth Generation Chipset (P5):

- Intel introduced a chipset called 430LX with the introduction of its first Pentium processor. Following that, Intel released different versions of this chipset at different times. Some are: 430LX (Mercury), 430NX (Neptune), 430FX (Triton), 430MX, 430 HX, 430 VX, and 430 TX.

Sixth Generation Chipset (P6):

- Sixth generation chipsets are chipsets that work with Pentium pro/II/III. The chipsets are of two architectures: hub architecture and north and south bridge architecture.

Pentium Pro chipsets are:

- North and south bridge: 450KX (Orion), 450 GX(Orion server), 440FX(Natoma).
- Hub architecture: 810(Whitney), 810E (Whitney), 820(Camino), 820E, 840

Intel 810:

- Introduced in 1999. This chipset represents a major change in chipset design from standard North South Bridge to hub architecture design. It allows for improvement in system performance all for less cost and system complexity.

Seventh Generation Chipset (P7):

- With the introduction of P IV, Intel introduced new chipset numbered as 9XX. These include: 910GML, 910GL, 915P, 925X, 925XE, 955X, 975X,etc. All of them use hub architecture design.

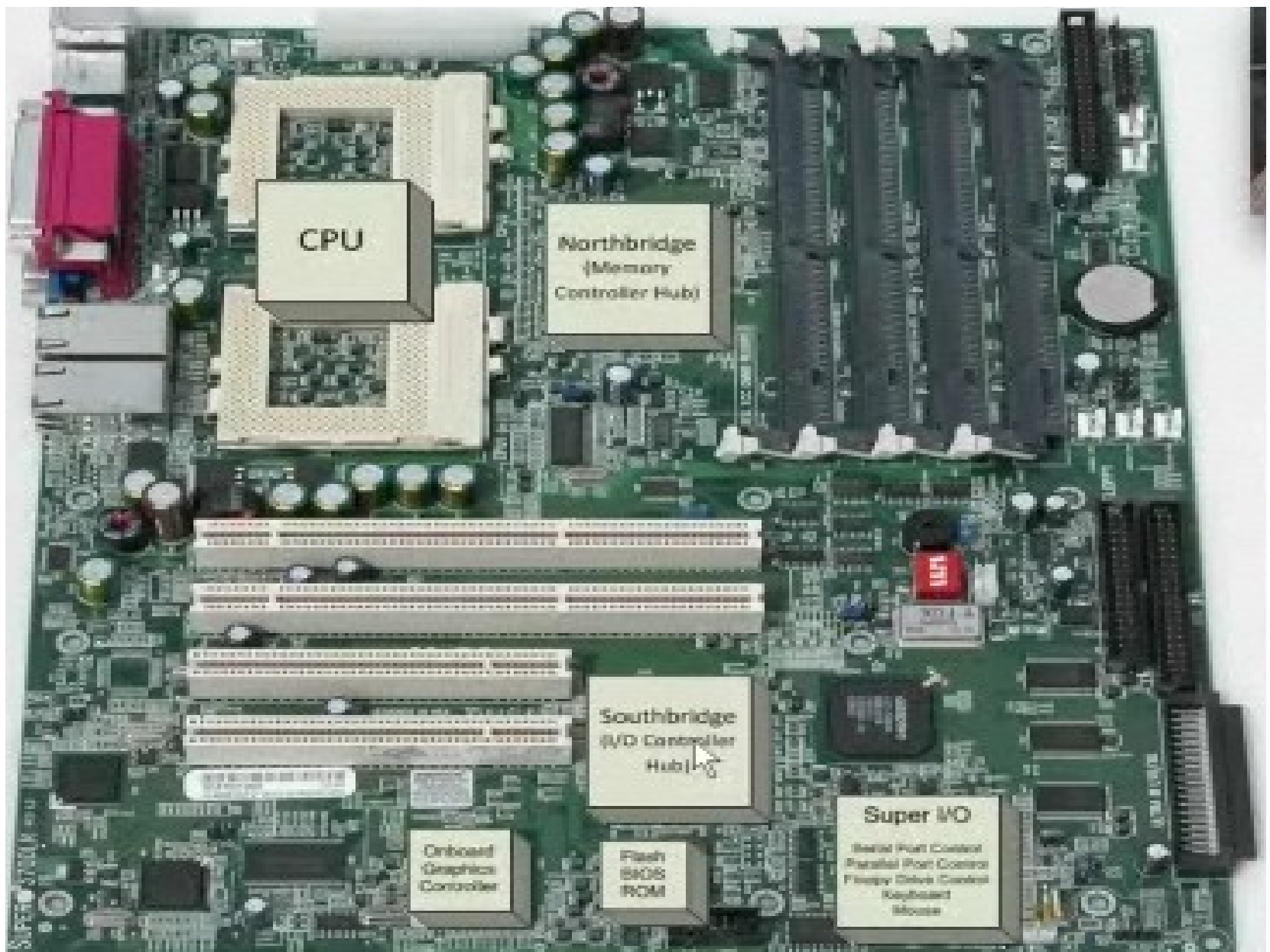


Fig the motherboard and chipsets

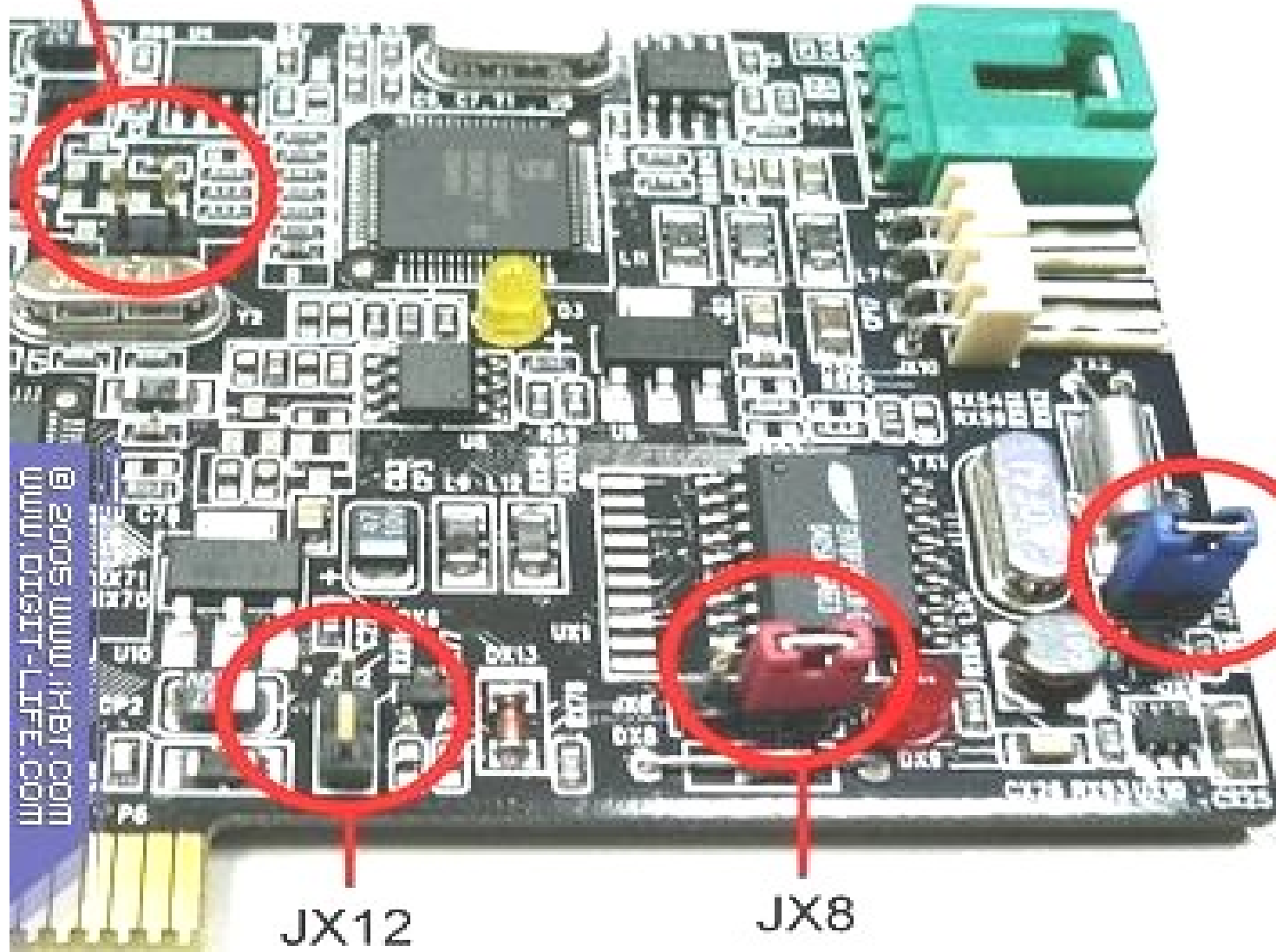
Motherboard Configuration

- Motherboard settings can be configured in three ways:
 - Using jumpers
 - Using DIP switches – a tiny switch or group of switches on a circuit board. Named for the form factor of the carrier device in which the switch is housed.
 - Using BIOS

Jumpers

- Jumpers are pins on a motherboard or other device that are used to provide configuration information to the hardware.
- A single jumper consists of a pair of pins, with a small rectangular shunt that can be placed over both pins to short them together.
- The hardware is programmed to act one way when the jumper is shorted, and another way when it is left open. The jumpers are normally numbered JP1, JP2 etc.

JX9



JX12

JX8

JX13

Fig Jumpers

- Every motherboard differs in its jumper numbering, positioning, and most importantly, what the settings for each jumper mean. This is why having the motherboard manual is so crucial for anyone who wants to work on their PC.
- One of the newest innovations on the market is the "jumperless" motherboard. With these boards you make many of the hardware settings, such as CPU type and speed, and even CPU voltage, using BIOS settings, while some others are retained as regular jumpers (typically, CMOS clear and cache size).
- This arrangement lets you change the clock speed of the CPU as easily as you would change any other BIOS setting, and makes upgrading the processor a snap.
- It also makes it easier for motherboard manufacturers to add support for new processor types as they are put onto the market.

- Here is a list of the settings that you are most likely to set using jumpers:

- 1) **Processor/CPU Voltage:** Almost all newer boards have one or more jumpers to set the voltage for the processor. Obviously, this needs to be set correctly or you risk destroying the chip. In addition, newer processors use two voltages: an "external" I/O voltage and an "internal" core voltage. For your motherboard to support these newer processors, it must have jumpers to set both of these voltage types.
- 2) **Processor Speed / Bus Speed / Multiplier:** Every 486 class or later motherboard (excluding jumper less designs) has a way to specify the speed of the processor. There are two main methods for selecting processor speed, and your manual will show you which your motherboard uses. Some motherboards provide a list of the various speeds supported, and a diagram of how to set the jumpers to match that speed. Others instead require you to set two separate jumpers: one controls the memory bus speed, and the other sets the processor "multiplier" (how many times the memory bus speed the processor runs).

3) Processor Type: In an effort to save the user the hassle of figuring out voltages and speeds, some motherboards combine the jumpers into one jumper set or group.

- Then, they provide a long list with each processor type and speed that the board supports, and tell you how to set each of a group of jumpers so that it works appropriately.
- If you are setting up a Pentium 133, you'll find it on the list and be told how to set jumpers 1, 3, 7, 12, and 18, for example. The only problem with this arrangement is that unless they also tell you what each of the individual jumpers is actually controlling, it can be difficult to set up a newer processor that may be supported just fine by the board but not be listed in their documentation.

- 4) **Cache Size and Type:** Some boards can have different amounts of cache, and some can have cache either built in. There is often a jumper or two to set which is used and how much is on the board.
- 5) **Memory Size/Type:** Almost all newer PCs auto-detect the amount and type of RAM in the system, but many older 486 (and earlier) models require jumpers to be changed when changing the amount of memory.
- 6) **Flash BIOS Enable:** Many boards require you to set a jumper to a special position in order to enable the flash BIOS update feature.
- 7) **CMOS Clear:** Some newer PCs have built into the motherboard the ability to clear the BIOS by temporarily setting a jumper and then replacing it (in essence this disconnects the CMOS memory from the battery so it discharges and resets).
- 8) **Battery Source:** Some boards let you switch the battery from an internal to an external source, and a jumper is used to control the setting.

- **Configuration Procedure**

- This procedure describes the steps required to configure a motherboard for installation. You will normally use this procedure when building a new PC or performing a motherboard upgrade. Configuring the motherboard is traditionally done using jumpers.

- **Steps:**

- 1) **Read motherboard manual:** Trying to configure a motherboard without having the manual around is dangerous. Read the section that describes the jumpers on your particular board. Look at the picture of the board included in most decent manuals, to physically locate where the jumpers are.
- 2) **Set Processor Voltage Jumper(s):** Most newer motherboards will have two voltage specifications, one for the internal (core) voltage of the CPU, and one for the external (I/O) voltage. Be sure to set the correct levels for your CPU; again, the manual may tell you what settings you need for the processors it supports. If you are using an older processor without split-rail voltage, set both settings the same.
- 3) **Set Processor Speed Jumpers:** The speed of the processor is determined by two primary jumper settings: the system bus speed, and the multiplier (how many times the bus speed the CPU speed is). For example, a Pentium with MMX 200 uses a system bus speed of 66 MHz and a multiplier of 3x. Watch out for newer CPUs that use a lower multiplier which is interpreted differently by the CPU. For example, 233 MHz processors are often jumpered as 66x1.5, and the CPU interprets the 1.5x as 3.5x internally. Again, consult your manual.

- 4) **Set Secondary Cache Size and Type Jumper:** Some motherboards accept cache in different configurations and therefore have jumpers depending on how much cache is on the board and whether it is soldered on or inserted via a cache module.
- 5) **Check Flash BIOS Jumper:** If your system has a jumper to enable the flash BIOS feature, check it to make sure that it is disabled. This should be the default.
- 6) **Check CMOS Clear Jumper:** Some systems have a jumper that will let you clear the contents of the CMOS memory, something that is needed most often when a system password is set and then forgotten. Make sure that this jumper is set to the normal or default position, or you won't be able to set any BIOS settings.
- 7) **Check Battery Source Jumper:** Some motherboards use a jumper to determine if the onboard battery is to power the CMOS memory, or an external battery. Again, make sure this is set to the default position (onboard battery) unless using an external.
- 8) **Check Disable Jumpers:** Some motherboards have special jumpers to allow you to enable or disable parts of the motherboard at a hardware level (for example, the serial/parallel ports or the floppy disk controller). Make sure that these jumpers are set properly (normally, you won't want to disable any of these items).
- 9) **Set Memory Size Jumpers:** Very rarely seen on new boards, jumpers to set the size of the system memory were common on 486-class boards. Set these if your board requires them.

- **Troubleshooting Motherboard**

- Here are lists of common problems of motherboard:

1) When you shutdown the PC, it loses date, time, and BIOS settings.

- Solution: Change CMOS battery. It is exhausted.

2) Your mouse or keyboard does not work.

- These ports are not built to withstand repeated insertion. The solution is to resolder them or have professional resolder fix them.

3) The system does not work after installing new components:

- Possible Solutions:

- Check if the CPU, RAM, or AGP card is properly installed or fit
- Remove the new component and run the system
- Check the system for some sort of burn during the process
- Check the jumper setting of the system.

- Tips to troubleshoot motherboard:
 - 1) Remove peripheral components from motherboard including expansion cards, external peripherals, etc. and see if the problem is solved
 - 2) Check the motherboard jumper setting: make sure that they are all correct and not causing the problem.
 - 3) Reset BIOS setting to default and check
 - 4) Inspect the motherboard for:
 - Is Motherboard cracked?
 - Is there any broken pin?
 - Is power Connector fit properly?
 - Is there any loose socketed components?
 - Are all cables connected properly?